Computer Engineering
Machine Learning — Regression (Neural Network)

Name:_______________________ **Date:** _______________________
Year & Section: BSCpE 3-A
Course: Computer Engineering — Embedded Systems

General Instructions

Answer ALL THREE QUESTIONS below. Each question is an independent regression problem requiring you to apply the full six-step ML pipeline (preprocessing → normalization → model creation → training → validation → testing). Each question is worth 100 points; the exam totals 300 points.

Dataset Usage Instructions

1. **Train / Test Split (30:10 hold-out).** Use the first 30 rows (IDs ending in 001–030) as the training set, and the last 10 rows (IDs ending in 031–040) as the test set. Do not shuffle before splitting — rows are already arranged so both subsets contain a representative mix of low / mid / high target values.
2. **Categorical Encoding.** Apply one-hot encoding to every string column. Fit the encoder only on the training rows, then transform both sets so that no test-set information leaks into training.
3. **Feature Scaling.** Standardize the numeric features (z-score: mean 0, std 1) using a scaler fit only on the training rows, then transform the test rows with the same scaler.
4. **Model Setup (Feed-Forward Neural Network — Regressor).** Input layer = total encoded + scaled features; 1–2 hidden dense layers (8–16 neurons, ReLU); output layer = 1 neuron, linear activation. Loss: MSE. Optimizer: Adam (lr ≈ 0.001). Train 200–500 epochs, batch size ≈ 8.
5. **Evaluation on the Test Set.** Compute MAE (average absolute error), RMSE (penalizes large errors), and R^2 (closer to 1.0 is better). Report a predicted-vs-actual comparison table.
6. **(Optional) Cross-Validation.** For hyperparameter tuning, apply 5-fold CV only on the 30 training rows. Never touch the test rows during tuning — they are evaluated exactly once, at the end.

Question 1 — Rooftop Solar Inverter Output

Scenario

A renewable-energy startup installs grid-tied rooftop solar systems across Metro Manila and wants to forecast each system's daily energy yield to validate Service Level Agreements with its clients. Field engineers log irradiance, weather, panel brand, panel tilt, panel age, dust accumulation, and the day's overall sky condition from on-site sensors and observers, then compare against the inverter's recorded kilowatt-hour output. They want a model that, given today's environmental readings and panel state, predicts how many kWh the array will deliver by sunset so the company can flag underperforming installations early.

Given — Combined Dataset (40 rows: 30 train + 10 test):

Day ID	Weather	Panel Brand	Irradiance (W/m²)	Temp (°C)	Tilt (°)	Age (mo)	Dust (g/m²)	Output (kWh)
D-001	Partly Cloudy	SolarX	720	32	15	6	1.2	28.4
D-002	Cloudy	VoltMax	580	29	15	18	3.5	21.8
D-003	Sunny	SolarX	810	34	20	3	0.8	32.1
D-004	Overcast	GreenRay	430	27	15	36	6.0	14.7
D-005	Partly Cloudy	VoltMax	650	31	25	12	2.1	24.9
D-006	Sunny	SolarX	760	33	20	8	1.5	30.2
D-007	Cloudy	GreenRay	510	28	15	24	4.8	18.3
D-008	Sunny	SolarX	890	35	25	2	0.5	34.7
D-009	Overcast	GreenRay	380	26	10	48	7.2	11.5
D-010	Partly Cloudy	VoltMax	690	31	20	14	2.8	26.0
D-011	Cloudy	VoltMax	540	30	15	30	5.1	19.6
D-012	Sunny	SolarX	770	33	25	5	1.0	31.0
D-013	Overcast	GreenRay	460	27	15	42	6.7	14.2
D-014	Sunny	SolarX	820	34	20	10	1.8	31.8
D-015	Hazy	VoltMax	610	30	20	20	3.2	23.4
D-016	Overcast	GreenRay	350	25	10	54	7.8	10.2
D-017	Partly Cloudy	SolarX	740	32	25	7	1.3	29.5
D-018	Cloudy	VoltMax	590	29	15	22	3.9	21.2
D-019	Sunny	SolarX	870	35	20	4	0.7	33.6
D-020	Cloudy	GreenRay	470	28	15	38	5.8	15.4
D-021	Partly Cloudy	VoltMax	700	32	20	16	2.5	26.8
D-022	Hazy	VoltMax	560	30	25	26	4.2	20.5
D-023	Sunny	SolarX	830	34	25	6	1.1	32.5
D-024	Overcast	GreenRay	410	26	10	50	7.5	12.8
D-025	Partly Cloudy	VoltMax	680	31	20	13	2.4	25.7
D-026	Hazy	VoltMax	520	29	15	28	4.5	18.9
D-027	Sunny	SolarX	790	33	25	9	1.6	30.7
D-028	Overcast	GreenRay	440	27	15	44	6.3	13.6
D-029	Partly Cloudy	SolarX	750	32	20	11	2.0	28.9
D-030	Partly Cloudy	VoltMax	630	30	25	19	3.0	24.1
D-031	Sunny	SolarX	800	33	20	4	0.6	32.0
D-032	Cloudy	VoltMax	550	29	15	26	4.0	20.0
D-033	Overcast	GreenRay	390	26	10	52	7.4	11.0
D-034	Partly Cloudy	SolarX	730	32	25	5	1.0	29.1
D-035	Hazy	VoltMax	600	30	20	25	3.5	22.4
D-036	Sunny	VoltMax	780	33	25	11	1.7	30.0
D-037	Cloudy	GreenRay	490	28	15	32	5.2	17.0
D-038	Sunny	SolarX	850	34	25	3	0.6	33.5
D-039	Overcast	GreenRay	420	27	10	46	6.8	13.4
D-040	Partly Cloudy	VoltMax	670	31	20	15	2.6	25.5

Data types: Day ID, Weather, Panel Brand (string / categorical); Tilt (integer °); Age (integer months); all other features continuous floats.

Required Model:

Regression — feed-forward Neural Network with one linear output neuron.

The target variable, Daily Energy Output (kWh), is a continuous numerical value whose relationship to irradiance, temperature, orientation, age, dust, sky condition, and manufacturer is non-linear and interactive. A neural-network regressor — with Weather Condition and Panel Brand one-hot encoded — learns these joint dependencies better than a simple linear model.

Input features: Weather Condition, Panel Brand, Avg Solar Irradiance, Ambient Temp, Panel Tilt Angle, Panel Age, Dust Accumulation.
Expected output: Predicted Daily Energy Output (kWh) — a continuous numerical value.

Question 2 — Electric Vehicle Remaining Range

Scenario

A local EV manufacturer is improving the range-estimation algorithm shown on its dashboard, because customers complain the "kilometers remaining" gauge fluctuates wildly under different driving conditions. Telemetry from fleet test drives records each car's selected driving mode, road type, battery state of charge, ambient temperature, air-conditioner load, elevation gain, and average driving speed, alongside the actual distance the car travels before the battery empties. The engineering team wants a data-driven model that predicts the realistic remaining range in kilometers given the car's current operating context.

Given — Combined Dataset (40 rows: 30 train + 10 test):

Trip ID	Mode	Road Type	SoC (%)	Temp (°C)	AC (kW)	Elev (m)	Speed (km/h)	Range (km)
T-001	Eco	Urban	80	28	1.2	50	55	232
T-002	Sport	Highway	50	35	2.5	120	70	118
T-003	Eco	Urban	95	24	0.8	20	50	305
T-004	Sport	Mountain	30	33	2.0	200	80	58
T-005	Normal	Mixed	65	30	1.5	80	60	175
T-006	Eco	Urban	85	26	1.0	40	55	258
T-007	Sport	Mountain	45	34	2.2	150	75	92
T-008	Eco	Urban	90	25	0.9	30	50	285
T-009	Sport	Mountain	25	36	2.8	220	85	42
T-010	Normal	Mixed	70	29	1.4	70	60	198
T-011	Sport	Highway	55	32	1.8	110	70	130
T-012	Eco	Urban	88	27	1.1	45	55	265
T-013	Sport	Mountain	40	33	2.3	170	75	78
T-014	Normal	Mixed	75	28	1.3	65	60	215
T-015	Normal	Highway	60	31	1.7	100	65	158
T-016	Sport	Mountain	35	35	2.6	190	80	65
T-017	Eco	Urban	92	26	1.0	35	50	295
T-018	Normal	Mixed	58	30	1.6	90	65	152
T-019	Normal	Mixed	78	29	1.3	60	58	222
T-020	Sport	Mountain	48	34	2.4	160	75	88
T-021	Eco	Urban	82	27	1.1	55	55	244
T-022	Sport	Mountain	38	33	2.5	180	78	70
T-023	Normal	Mixed	68	30	1.5	75	62	188
T-024	Sport	Mountain	28	36	2.7	210	82	48
T-025	Eco	Urban	87	26	1.0	38	52	272
T-026	Sport	Highway	52	32	1.9	130	72	105
T-027	Normal	Mixed	72	29	1.4	68	60	205
T-028	Sport	Mountain	43	34	2.3	165	76	82
T-029	Normal	Highway	63	31	1.7	95	65	165
T-030	Sport	Mountain	33	35	2.6	195	80	60
T-031	Eco	Urban	83	27	1.1	42	53	245
T-032	Sport	Mountain	32	35	2.7	205	81	55
T-033	Normal	Mixed	67	30	1.5	78	61	182
T-034	Eco	Urban	91	25	0.9	28	51	290

Trip ID	Mode	Road Type	SoC (%)	Temp (°C)	AC (kW)	Elev (m)	Speed (km/h)	Range (km)
T-035	Sport	Highway	53	32	1.9	125	71	112
T-036	Normal	Highway	62	31	1.7	98	65	160
T-037	Sport	Mountain	37	34	2.4	175	77	72
T-038	Eco	Urban	89	26	1.0	36	53	278
T-039	Normal	Mixed	74	28	1.3	67	60	212
T-040	Sport	Mountain	27	36	2.8	215	83	45

Data types: Trip ID, Driving Mode, Road Type (string / categorical); Battery SoC (integer %); Elevation Gain (integer meters); all other features continuous floats.

Required Model:

Regression — feed-forward Neural Network with one linear output neuron.

The target variable, Remaining Range, is a continuous distance measurement (km) influenced non-linearly by driving mode, road type, battery state, climate, accessory load, terrain, and speed. A neural-network regressor — with Driving Mode and Road Type one-hot encoded — models these compounded effects more accurately than a static lookup table.

Input features: Driving Mode, Road Type, Battery SoC, Ambient Temp, AC Power Draw, Elevation Gain, Avg Speed.
Expected output: Predicted Remaining Range (km) — a continuous numerical value.

Question 3 — Data-Center Server Response Latency

Scenario

A cloud-hosting provider is tuning its auto-scaling rules to keep average API response times under its 100-millisecond SLA. Operations engineers log per-minute snapshots of each server's region, request type profile, CPU utilization, active connections, cache hit rate, available network bandwidth, and request queue depth, paired with the measured mean response latency for that minute. They want a predictive model that, given the current operating conditions, estimates the expected latency in milliseconds so the orchestrator can pre-emptively spin up additional instances before the SLA is breached.

Given — Combined Dataset (40 rows: 30 train + 10 test):

Snap ID	Request Type	Region	CPU (%)	Conns	Cache (%)	BW (Mbps)	Queue	Latency (ms)
S-001	Read-Heavy	US-East	45	320	92	850	12	38
S-002	Mixed	Asia-Pacific	78	1100	65	320	85	142
S-003	Read-Heavy	EU-West	30	180	95	940	5	22
S-004	Write-Heavy	Latin-America	88	1500	58	210	130	198
S-005	Mixed	US-East	62	720	81	600	40	76
S-006	Read-Heavy	EU-West	52	420	88	780	18	48
S-007	Write-Heavy	Asia-Pacific	82	1280	60	280	110	175
S-008	Read-Heavy	US-East	38	240	93	880	8	28
S-009	Write-Heavy	Latin-America	92	1620	55	180	145	215
S-010	Mixed	EU-West	68	850	78	540	52	92
S-011	Read-Heavy	US-East	48	380	89	820	15	42
S-012	Mixed	Asia-Pacific	75	1020	70	380	78	128
S-013	Read-Heavy	EU-West	34	210	94	920	6	25
S-014	Write-Heavy	Latin-America	85	1380	62	250	120	188
S-015	Mixed	US-East	58	560	84	690	30	65
S-016	Write-Heavy	Latin-America	95	1750	52	160	158	232
S-017	Read-Heavy	EU-West	42	290	91	800	11	35
S-018	Mixed	Asia-Pacific	72	920	74	460	65	108
S-019	Read-Heavy	US-East	50	460	87	740	22	52
S-020	Write-Heavy	Latin-America	80	1180	64	300	95	162
S-021	Read-Heavy	EU-West	36	220	94	900	7	26
S-022	Mixed	Asia-Pacific	65	780	79	580	45	84
S-023	Write-Heavy	Latin-America	90	1540	56	200	138	208
S-024	Mixed	US-East	55	510	85	660	26	58
S-025	Mixed	Asia-Pacific	70	880	76	500	58	98
S-026	Read-Heavy	EU-West	45	350	90	810	14	40
S-027	Write-Heavy	Latin-America	86	1440	59	230	125	195
S-028	Mixed	US-East	60	660	82	620	36	72
S-029	Read-Heavy	EU-West	32	200	96	930	4	23
S-030	Mixed	Asia-Pacific	77	1080	68	360	82	135
S-031	Read-Heavy	US-East	40	280	92	830	10	32
S-032	Write-Heavy	Latin-America	87	1460	60	220	132	200
S-033	Mixed	Asia-Pacific	73	970	72	420	70	118
S-034	Read-Heavy	EU-West	33	200	95	910	5	24

Snap ID	Request Type	Region	CPU (%)	Conns	Cache (%)	BW (Mbps)	Queue	Latency (ms)
S-035	Mixed	US-East	57	540	84	680	28	62
S-036	Write-Heavy	Latin-America	93	1680	54	170	152	225
S-037	Read-Heavy	EU-West	47	360	90	830	13	41
S-038	Mixed	Asia-Pacific	66	810	78	560	48	88
S-039	Write-Heavy	US-East	84	1320	61	260	115	180
S-040	Mixed	EU-West	53	470	86	700	24	55

Data types: Snapshot ID, Request Type, Server Region (string / categorical); all numeric inputs are integers; Response Latency is a continuous float.

Required Model:

Regression — feed-forward Neural Network with one linear output neuron.

The target variable, Response Latency (ms), is a continuous numerical quantity whose dependence on request type, region, CPU load, traffic, cache efficiency, bandwidth, and queue backlog is highly non-linear — small individual changes can compound into latency spikes. A neural-network regressor — with Request Type and Server Region one-hot encoded — captures these interactions, enabling proactive scaling.

Input features: Request Type, Server Region, CPU Utilization, Active Connections, Cache Hit Rate, Available Bandwidth, Queue Depth.
Expected output: Predicted Response Latency (ms) — a continuous numerical value.

Grading Rubric (per question — apply to Q1, Q2, Q3)

#	Criterion	Max	Q1 Score	Q2 Score	Q3 Score
1	Data Preprocessing	**15**	_____ / 15	_____ / 15	_____ / 15
2	Normalization	**10**	_____ / 10	_____ / 10	_____ / 10
3	Model Creation	**20**	_____ / 20	_____ / 20	_____ / 20
4	Training	**20**	_____ / 20	_____ / 20	_____ / 20
5	Validation	**15**	_____ / 15	_____ / 15	_____ / 15
6	Testing	**20**	_____ / 20	_____ / 20	_____ / 20
	Per-question subtotal	**100**	**_____ / 100**	**_____ / 100**	**_____ / 100**
	GRAND TOTAL (Q1 + Q2 + Q3)	**_____ / 300**			

Detailed Performance Bands per Criterion

1. Data Preprocessing (15)	13–15: all steps correct, no leakage. 9–12: minor mistake (e.g., encoder fit on full set). 5–8: split wrong or encoding incomplete. 0–4: missing or major leakage.
2. Normalization (10)	9–10: scaler fit only on training; same transform applied to test. 6–8: scaler fit on whole set. 3–5: inconsistent. 0–2: none.
3. Model Creation (20)	17–20: correct NN architecture + MSE/Adam + justification. 13–16: working but missing justification or minor issue. 8–12: structural error. 0–7: no working model.
4. Training (20)	17–20: trains cleanly, loss converges, curve shown, no overfitting. 13–16: minor issues. 8–12: no convergence or severe overfitting. 0–7: training fails.
5. Validation (15)	13–15: proper held-out validation or K-fold CV, used for tuning. 9–12: validation done but unused. 5–8: incomplete or test used as validation. 0–4: none.

6. Testing (20)	17–20: MAE/RMSE/R^2 reported + predicted-vs-actual + interpretation. 13–16: shallow interpretation. 8–12: only one metric or error. 0–7: no evaluation.

Deliverables (What to Submit)

1. Source code (Python notebook .ipynb or .py) implementing all six steps in order for each of the three questions.
2. Output report (3–6 pages) containing, for EACH question: preprocessing summary, model architecture, training loss curve, validation results, final test metrics (MAE / RMSE / R^2), and a 3–4-sentence business interpretation.
3. Clearly label each section of the submission as Q1, Q2, or Q3 so the grader can map your work to the correct question.

Submission Notes

- Late submissions: −10 pts per day.
- Plagiarized or shared code (between students assigned the same version): both submissions receive zero
- .
- Use of any library (scikit-learn, TensorFlow / Keras, PyTorch) is permitted as long as the six-step structure is followed.

Instructor Comments

Section	Notes / Feedback
Data Preprocessing	
Normalization	
Model Creation	
Training	
Validation	
Testing	
Overall	

Grader's Signature: ______________________________ Date Graded: _______________

Final Grade Equivalent: ☐ A (90–100) ☐ B (80–89) ☐ C (70–79) ☐ D (60–69) ☐ F (<60)